Phase 2 — Grade 2: Calculus & Gradients

Detailed Revision Guide — full formula breakdowns, edge cases, and common-mistake debugging

1. Derivatives — The Core Idea

A **derivative** answers one question: 'if I nudge the input a tiny bit, how much does the output change?' It is the INSTANTANEOUS rate of change at one exact point -- not an average over a range.

Analogy that makes it click: you're driving a car. Position changes over time. Your SPEED at any instant is the derivative of position with respect to time -- 'how fast is distance changing right now.' If you're stopped, speed (the derivative) is 0.

Notation: f'(x) (read 'f prime of x') and df/dx mean EXACTLY the same thing: 'the rate of change of f with respect to x.' The prime mark (') is just shorthand for 'the derivative of'.

1.1 The Power Rule

Formula

For $f(x) = x^n$, $f'(x) = n \cdot x^{(n-1)}$

In words: bring the exponent DOWN as a multiplier, then REDUCE the exponent by 1.

Where do n and x come from? **n is whatever exponent is already written in the function** -- you read it off, you don't choose it. **x stays symbolic** (a placeholder, not a number) unless the problem separately asks you to evaluate the derivative at a specific point -- and that substitution is ALWAYS a separate, later step, never done while you're still finding the general formula.

Worked Example 1

$f(x) = x^3$ (here n=3)

$f'(x) = 3 \cdot x^{(3-1)} = 3x^2$

Now evaluate at x=3 (a SEPARATE step): $f'(3) = 3 \cdot 3^2 = 3 \cdot 9 = 27$

COMMON MISTAKE: Doing the arithmetic too fast at the final multiplication
A real example from practice: f'(3) = 3×9 was miscalculated as 21 instead of 27. The RULE was applied perfectly ($3 \cdot x^2$, then substitute x=3) -- the slip was purely in the final multiplication step. **Debug habit:** after finishing a multi-step formula, do the LAST arithmetic step slowly and separately, double-checking it on its own before accepting the final answer.

1.2 The $x^0=1$ Edge Case (n=1 in the Power Rule)

Formula

$f(x) = x$ (same as x^1, so n=1)

$f'(x) = 1 \cdot x^{(1-1)} = 1 \cdot x^0$

RULE: any nonzero number raised to the power 0 equals 1 (not the number itself)

f'(x) = 1 · 1 = 1

EDGE CASE: f(x)=x has a CONSTANT derivative, not a variable one
f'(x) = 1 for EVERY value of x -- it does not depend on x at all. This makes sense because f(x)=x is a perfectly straight 45-degree line: every 1-unit nudge in input produces exactly a 1-unit change in output, everywhere on the line, so the rate of change never varies.

COMMON MISTAKE: Writing f'(x)=x instead of f'(x)=1
A real example from practice: for f(x)=x, the answer 'f'(x) = x' was given at first. This happens when you forget to actually APPLY the power rule and instead just copy the original function. The fix: always write out x^0 explicitly as an intermediate step, then immediately convert it to 1 using the x^0=1 rule, so you never skip past it.

1.3 The Constant Rule

Formula

If f(x) = c (c is a plain number, no x anywhere), then f'(x) = 0

Why this rule matters (not just 'it's always zero'): constants show up ATTACHED to real functions, e.g. f(x) = x^2 + 100. The 100 could represent a fixed baseline/offset/intercept (like the 'b' in y=mx+b). The constant rule is what lets you correctly say that attached constant contributes NOTHING to the rate of change, and can be dropped to 0 -- without this rule you might wrongly assume the constant has to do something in the derivative.

COMMON MISTAKE: Applying the constant rule to the WRONG thing
A real example from practice: differentiating f(x) = x^2 + x, the first term correctly became 2x, but the second term's derivative (x → 1) was then INCORRECTLY treated as a constant and dropped to 0, giving the wrong final answer '2x' instead of the correct '2x + 1'. The constant rule only applies to a constant sitting in the ORIGINAL function BEFORE differentiation -- it does not apply to a derivative RESULT just because that result happens to be a plain number (like the '1' from differentiating x). The sum rule says: differentiate each term, then ADD all the results -- nothing found during differentiation gets thrown away afterward.

1.4 The Sum Rule & Constant-Multiplier Handling

Formula

If f(x) = g(x) + h(x), then f'(x) = g'(x) + h'(x) (differentiate each term SEPARATELY, then add)

A constant MULTIPLYING an x-term stays in place; only the power rule applies to the x-part:

$5x^2$ -> 5 · (2x) = 10x

Worked Example (all rules combined)

f(x) = $2x^2$ + 3x + 10

Term 1: $2x^2$ -> 2 · 2x = 4x

Term 2: 3x -> 3 · 1 = 3 (recall: derivative of x alone is 1)

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Term 3: 10 -> 0 (constant rule)
f'(x) = 4x + 3 + 0 = 4x + 3 <-- GENERAL FORMULA, found FIRST
Only now, as a separate step: f'(2) = 4(2) + 3 = 8 + 3 = 11
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COMMON MISTAKE: Substituting a number BEFORE finishing the general formula
A real example from practice: when asked to find f'(x) then evaluate f'(2), the number 2 was plugged in midway through differentiating, mixing both steps into one tangled line. **Debug habit:** ALWAYS write the complete symbolic f'(x) formula on its own line first, with no numbers substituted. Only on the NEXT line, substitute the specific value. Never combine these two steps into one line of working.

1.5 Full Derivation: Why the Power Rule Formula is TRUE (not just a rule to memorize)

This connects the formula back to a real numeric calculation, so it feels earned rather than arbitrary.

Step-by-step derivation for $f(x) = x^2$

Compare point x to a nearby point x+h (h = a tiny nudge)

Step 1 -- expand: $f(x+h) = (x+h)^2 = x^2 + 2xh + h^2$

Step 2 -- average rate of change over the gap h:

$[f(x+h) - f(x)] / h = [x^2+2xh+h^2 - x^2] / h = [2xh+h^2]/h$

Step 3 -- cancel h from every term: = 2x + h

Step 4 -- check against real numbers (x=2, h=0.01): 2(2)+0.01 = 4.01 <- matches a direct numeric check

Step 5 -- the derivative is what this expression approaches as h shrinks toward 0:

as h -> 0: 2x + h -> 2x => f'(x) = 2x (now actually DERIVED, not just stated)

EDGE CASE: Average rate of change works for ANY gap size, not just tiny ones
The formula [f(b)-f(a)]/(b-a) is valid for a huge gap too (e.g. x=2 to x=5), not only a tiny nudge. For $f(x)=x^2$: f(2)=4, f(5)=25, average rate = (25-4)/(5-2) = 21/3 = 7. This IS a valid number (like an average speed over a long trip) but it BLENDS the whole interval together -- it does NOT tell you the instantaneous rate at x=2, at x=5, or anywhere between, since the true instantaneous rate (f'(x)=2x) is actually different at every single point in that range. Small gap -> close approximation of ONE point's instantaneous rate. Large gap -> a valid but coarser, blended average.

TIP: Average vs. Instantaneous -- the one-line distinction to memorize
Average rate of change = change measured over a GAP/interval between two points (an approximation unless the gap is zero). Instantaneous rate of change = the EXACT rate at one single point, with no gap at all -- this is what the derivative gives you. The average becomes equal to the instantaneous rate only in the limit as the gap shrinks all the way to zero.

2. Partial Derivatives & Gradients

Real functions often depend on MULTIPLE variables at once (e.g. price depends on both size AND location). A **partial derivative** asks: nudge ONE variable, hold every other variable completely frozen/constant, how does the output change? Do this separately for each variable, then bundle all the results into one vector -- that vector is called the **gradient**.

Analogy that makes it click (the hillside): standing on a hill, altitude = f(x,y), where x=east-west position and y=north-south position. Walking east changes altitude at ONE rate (∂f/∂x); walking north changes it at a genuinely DIFFERENT rate (∂f/∂y). A single number can't capture the hill's steepness, because steepness truly differs by direction -- that's exactly why the gradient bundles two (or more) separate rates together instead of being one number.

2.1 The Mechanics: Why the Other Variable Becomes a Constant

Take the simplest possible case: f(x,y) = x + y. To find ∂f/∂x, FREEZE y at some fixed number -- say y=10 (the actual number doesn't matter). The function becomes f(x) = x + 10, which is now an ORDINARY one-variable function. The frozen 10 is just a constant, and by the constant rule its derivative is 0. So ∂f/∂x = 1 + 0 = 1. It doesn't matter WHAT number y was frozen at -- a frozen constant always differentiates to 0 regardless of its actual value, which is why we can just say 'treat y as a constant' without ever picking a specific number for it.

Worked Example: f(x,y) = $x^2 + y^2$

∂f/∂x (freeze y): x^2 -> 2x; y^2 (frozen/constant) -> 0

∂f/∂x = 2x + 0 = 2x

∂f/∂y (freeze x): x^2 (frozen/constant) -> 0; y^2 -> 2y

∂f/∂y = 0 + 2y = 2y

Gradient: ∇f = [∂f/∂x, ∂f/∂y] = [2x, 2y]

Numeric verification at x=3, y=4 (the hillside, real numbers)

Nudge x only (y fixed at 4): x: 3 -> 3.01. f(3,4)=25, f(3.01,4)=25.0601. rate=(25.0601-25)/0.01=6.01

Matches 2x = 2(3) = 6 ✓

Nudge y only (x fixed at 3): y: 4 -> 4.01. f(3,4)=25, f(3,4.01)=25.0801. rate=(25.0801-25)/0.01=8.01

Matches 2y = 2(4) = 8 ✓

Worked Example 2: f(x,y) = $3x^2$ + 5y

∂f/∂x: $3x^2$ -> 3·2x = 6x; 5y (y treated as constant here, but this whole TERM contains no x) -> 0

∂f/∂x = 6x

∂f/∂y: $3x^2$ (contains no y) -> 0; 5y -> 5·1 = 5

∂f/∂y = 5

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Gradient: ∇f = [6x, 5] (notice: one entry stays symbolic, the other is a
plain constant -- both are valid in the same gradient vector)
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COMMON MISTAKE: Forgetting that a term with ONLY the frozen variable disappears completely
When finding $\partial f/\partial x$ for $f(x,y)=3x^2+5y$, it's easy to accidentally keep the '5y' term unchanged instead of recognizing that -- since y is frozen -- the ENTIRE term 5y is just 'some constant,' which differentiates to 0 by the constant rule. Any term that contains ONLY the variable being held constant (no trace of the variable you're differentiating with respect to) drops to 0 entirely, no matter how many pieces or coefficients it has.

EDGE CASE: A term with BOTH variables mixed together (not covered yet, but worth knowing)
All examples above kept x-only and y-only terms cleanly separate (like x^2+y^2). If a term mixes both variables together (e.g. xy), differentiating it requires the PRODUCT RULE, a rule not yet covered in this grade -- for $\partial(xy)/\partial x$, you would treat y as a constant multiplier, giving $\partial(xy)/\partial x = y$ (not 0, because x is still present). This is flagged here as a known gap to revisit if you encounter mixed terms before the product rule is formally taught.

3. The Chain Rule

Some functions are NESTED -- one function feeding into another, like $f(x) = (x^2+1)^3$. The chain rule handles this: differentiate the OUTER function (using a placeholder for the inner part), differentiate the INNER function alone, then MULTIPLY the two results together.

Analogy that makes it click (salary/rating/hours): salary depends on rating; rating depends on hours worked -- a two-link chain, hours → rating → salary. The TOTAL effect of hours on salary = (effect of rating on salary) × (effect of hours on rating). Each link's rate multiplies through the chain, like gears turning one another -- not like adding independent bricks.

3.1 The Numeric Proof (Why Multiply, Not Add)

Salary example with real numbers

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rating = hours × 2; salary = rating²

At hours=3: rating = 3×2 = 6; salary = 6² = 36

Link 1 (d(rating)/d(hours)) = 2 (flat multiplier: rating always changes 2x
as fast as hours)

Link 2 (d(salary)/d(rating)) = 2·rating = 2·6 = 12 (power rule, at rating=6)

Chain rule: d(salary)/d(hours) = Link2 × Link1 = 12 × 2 = 24

Verify with a direct nudge: hours=3.01 -> rating=6.02 -> salary=36.2404

rate = (36.2404-36)/0.01 = 24.04 ≈ 24 ✓ (matches the chain-rule
multiplication)
```

TIP: The multiplication IS the combination of two separate rates, nothing more
d(salary)/d(hours) = d(salary)/d(rating) × d(rating)/d(hours). This is literally multiplying two 'how much does Y change per unit of X' rates together to get the combined effect across both links -- the same pattern as unit conversion (km/hr × hr = km).

3.2 The 4-Step Placeholder Method (Full Breakdown)

The 4 steps, explained

STEP 1 -- NAME the inner function: give it a temporary label, u = g(x).

Example: for $f(x)=(x^2+1)^3$, let $u = x^2+1$, so $f = u^3$.

STEP 2 -- DIFFERENTIATE THE OUTER shape using the placeholder u (ordinary power rule, but on u instead of x):

$df/du = 3u^2$ <- do NOT substitute the real inner formula back in yet at this point

THEN substitute u's real value back in: $df/du = 3(x^2+1)^2$

STEP 3 -- DIFFERENTIATE THE INNER function ALONE: forget the outer function entirely. Just take the ordinary derivative of u's own formula, using rules already known (power/sum/constant rule).

$u = x^2+1 \rightarrow du/dx = 2x + 0 = 2x$

STEP 4 -- MULTIPLY Step 2's result by Step 3's result:

$f'(x) = df/du \times du/dx = 3(x^2+1)^2 \times 2x = 6x(x^2+1)^2$

Worked Example 2: $f(x) = (x+4)^2$

Step 1: $u = x+4$

Step 2: $f=u^2$ -> $df/du = 2u$ -> substitute back -> $2(x+4)$

Step 3: du/dx for $u=x+4$ -> $1 + 0 = 1$

Step 4: $f'(x) = 2(x+4) \times 1 = 2(x+4) = 2x+8$

EDGE CASE: When the inner function is LINEAR (like x+4 or x+5), Step 3 is always just the coefficient of x

If u = ax+b (any straight-line inner function), du/dx is always just 'a' -- a plain constant, with no x left in it. This means Step 4's multiplication just scales Step 2's result by that constant; it never introduces any NEW x-dependence of its own. This is worth recognizing as a shortcut-check: if the inner function is linear, the final answer's 'shape' (in terms of x) comes entirely from Step 2.

3.3 Debugging Walkthrough: Every Mistake Seen in Practice, in Order

COMMON MISTAKE: Merging 'the function' and 'its derivative' into one line (Step 2)

Example of the error: writing 'df/du = u^2 = $(x+5)^2$ = 2(x+5)' all on one line. The problem: u^2 IS f itself (the ORIGINAL function, before differentiating) -- it is not df/du. The correct sequence is: first apply the power rule to get df/du=2u (a NEW expression, different from u^2), and ONLY THEN substitute u's real value back in. Keep 'the function' and 'the derivative of the function' on clearly separate lines so they never get conflated.

COMMON MISTAKE: Confusing 'the function' with 'its derivative' (Step 3)
Example of the error: for u = x+5, writing 'du/dx = u = x+5 = x+0 = x' -- this just rewrites u with the constant dropped, which is NOT differentiation at all. du/dx asks for the DERIVATIVE of u, not a simplified rewrite of u. The correct process: apply the sum rule + constant rule PROPERLY -- x's derivative is 1 (not x itself), and 5's derivative is 0 (constant rule) -- giving du/dx = 1+0 = 1, a completely different (and much simpler) result than 'x'.

COMMON MISTAKE: Using leftover numbers from a PREVIOUS problem
Example of the error: after solving one problem with inner function x^2+1, the next problem (with inner function x+4) was accidentally solved using '$2u = 2x^2$' -- substituting back the WRONG formula for u, left over from the earlier problem, instead of the current problem's actual u=x+4. Debug habit: before substituting u's value back in, explicitly re-read Step 1's definition for THIS problem, not a remembered one from a previous problem.

COMMON MISTAKE: Forgetting to multiply Steps 2 and 3 together (Step 4)
The most common final-step error is stopping after Steps 2 and 3 are individually correct, without actually performing Step 4's multiplication -- leaving two separate correct pieces instead of one combined final answer. Always explicitly write 'f'(x) = [Step 2 result] × [Step 3 result]' as its own labeled line, even when Step 3's result is just the number 1 (multiplying by 1 is still a required, explicit step, not something to skip just because it doesn't change the value).

4. Full Debugging Checklist for This Grade

Run through this checklist on any derivative problem before finalizing an answer:

1. Have I written the GENERAL formula f'(x) completely, with no numbers substituted, before doing any substitution? 2. For every term with only a 'frozen'/constant variable (partial derivatives), did I drop it all the way to 0, not just leave it unchanged? 3. For x^1 or any 'x alone' term, did I correctly get 1 (using $x^0=1$), not 'x' itself? 4. For the chain rule, are Steps 2 and 3 clearly on SEPARATE lines, with Step 2 using the placeholder u before substituting back? 5. Did I actually perform the final multiplication (Step 4) rather than leaving two separate pieces? 6. Did I double-check the LAST arithmetic step (the final multiplication/addition) on its own, separately from the method?